Indian Institute of Technology Delhi

Trustworthy Large Language Model

ELL 8299

Assignment No.: 01
Fine-Tune, Sanitize, Extract

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Date of Submission: 17th September, 2026.

Part A - PEFT Adapter Sweep

A.1 Setup All six runs use Qwen2.5-1.5B-Instruct (approximately 1.54 B total parameters), 20,000 training / 2,000 validation examples from ai4privacy/pii-masking-300k, seed 42, one epoch, cosine learning rate with peak 2×10^{-4} and 10% warm-up, and effective batch size 16 (4 per-device x 4 gradient accumulation). The prompt is fixed and cross-entropy loss is computed only on assistant-turn tokens. Trainable-parameter counts are taken from model.print_trainable_parameters(), and peak GPU memory from torch.cuda.max_memory_allocated() after training.

A.2 LORA freezes the weight matrix:

$$W_0 \in \mathbb{R}^{d \times k}$$

and injects a trainable update:

$$W = W_0 + (\alpha/r)BA$$

where:

$$B \in \mathbb{R}^{d \times r} \text{ (init 0)}$$

$$A \in \mathbb{R}^{r \times k} \text{ (init } \mathcal{N}(0, \sigma^2))$$

$$r \ll \min(d, k)$$

Trainable params per matrix = r(d+k). QLoRA stores W_0 in 4-bit NF4 (information-optimal quantiles for $\mathcal{N}(0,1)$ weights), dequantising on-the-fly during the forward pass; adapters stay in bf16.

A.3 Sweep Results

Run	Method	Modules	r	alpha	Train. %	Peak GB	Time (s)	Val F1	LR (%)
A1	LORA	q.v	8	16	0.071	7.05	3,576.1	0.6830	3.77
A2	LORA	q,k,v,o	8	16	0.141	7.07	3,750.5	0.7296	3.77
A3	LORA	attn+MLP	8	16	0.595	7.18	5,195.6	0.7452	3.86
A4	LORA	attn + MLP	32	16	2.336	7.67	5,236.0	0.7309	4.47
A5*	LORA	attn+MLP	32	64	2.336	7.67	5,233.4	0.7504	3.68
A6	QLORA	attn+MLP	32	64	3.990	6.85	7,529.3	0.7474	3.77

Observation: It is noticed that A5 has the highest validation F1 in the current sweep (0.7504) and the lowest leak rate among A4-A6 (3.68%). Increasing rank from A3 to A4 while keeping alpha 16 lowers F1 from 0.7452 to 0.7309 because alpha/r falls from 2.0 to 0.5. Restoring alpha 64 in A5 raises F1 by 1.94 percentage points at the same rank and module set. A6 is close to A5 in F1 (0.7474 vs. 0.7504) while using 0.83 GB less peak GPU memory, a 10.8% reduction.

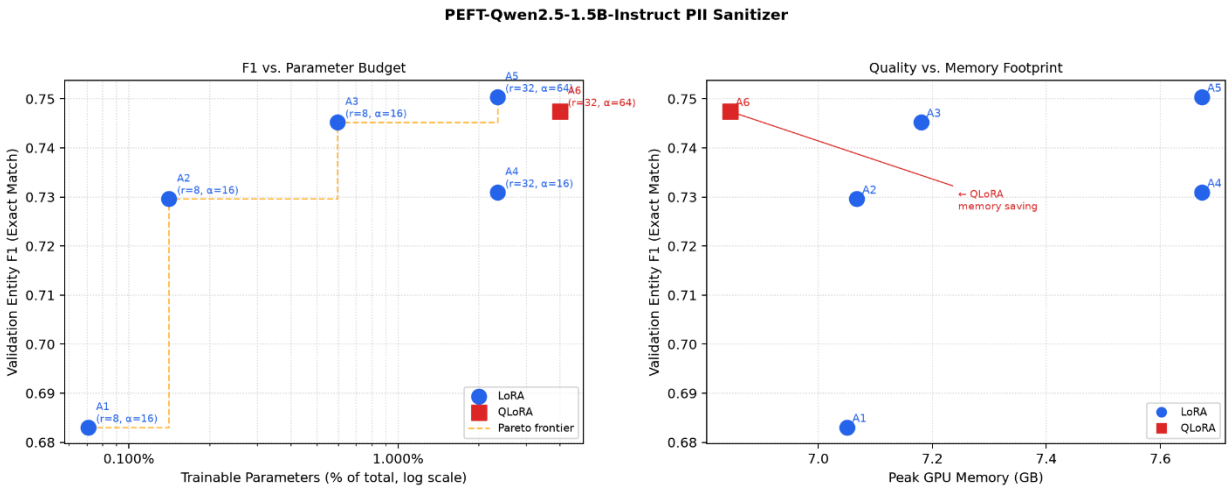


Figure 1.: A sweep visualization: F1 vs. trainable-parameter budget and validation quality vs. peak GPU memory.

Observation: The figure shows A1-A3 gaining F1 as more modules are adapted, A4 dipping after rank scaling without a matching alpha increase, and A5 recovering the loss through the alpha/r ratio. A6 remains close in quality to A5 with a lower memory footprint. These are descriptive relationships in this six-run sweep, not evidence that any single configuration is universally preferable.

Part B - Sanitizer Evaluation

B.1 Span Recovery The model emits plain masked text; spans are recovered using character-level SequenceMatcher (difflib, autojunk=False). For each 'replace' opcode, the predicted fragment is scanned for [LABEL] tokens using LABEL_RE `RE=r'

One label produces one span over the replaced source region; multiple labels are assigned by proportional splitting. Delete opcodes without a label are treated as UNKNOWN spans and therefore contribute false positives. Known failure modes include truncated output, adjacent PII merged into one label, label drift, and paraphrasing of non-PII text that creates spurious replace operations.

B.2-B.3 Metrics and Baselines

System	P (exact)	R (exact)	F1 (exact)	F1 (IoU>=0.5)	Leak Rate	Over-mask
Regex (ours)	0.482	0.237	0.318	0.322	54.92%	0.47%
Presidio default	0.184	0.184	0.183	0.184	0.195	47.29%
Qwen zero-shot	0.066	0.088	0.076	0.082	55.54%	24.56%
A5 fine-tuned	0.580	0.643	0.610	0.613	3.79%	0.46%

Observation: A5 has the highest exact micro-F1 in the current ID comparison (0.610) and the lowest leak rate (3.79%). Its exact recall (0.643) exceeds its precision (0.580), indicating that missed/incorrect spans remain a larger issue than excessive predictions under the exact criterion. Zero-shot has a much higher over-masking rate (24.56%) and a higher leak rate (55.54%) in the same export.

B.2a Per-Class Results - A5 fine-tuned

Label	Rec.	F1 exact	Prec.	F1 Iou	Support
TIME	0.675	0.742	0.823	0.745	945
USERNAME	0.853	0.826	0.840	0.841	760
SOCIALNUMBER	0.979	0.910	0.943	0.943	668
EMAIL	0.959	0.817	0.882	0.887	662
IDCARD	0.837	0.839	0.841	0.845	651
DRIVERLICENSE	0.585	0.816	0.456	0.589	643

Label	Prec.	Rec.	F1 exact	F1 IoU	Support
LASTNAME1	0.000	0.000	0.000	0.000	623
BOD	0.943	0.868	0.904	0.908	553
PASSPORT	0.894	0.734	0.806	0.810	552
SEX	0.975	0.970	0.973	0.976	529

Observation: Performance varies strongly by entity class. SEX (F1=0.973) and SOCIALNUMBER (0.943) are detected reliably, while DRIVERLICENSE recall is 0.456. LASTNAME1 has zero precision, recall, and F1 in the current export despite 623 gold spans, making it a dominant class-specific failure.

B.4 OOD Generalisation - A5

Metric	A5 OOD
Exact micro-P	0.080
Exact micro-R	0.108
Exact micro-F1	0.092
IoU>=0.5 micro-F1	0.101
Leak rate	34.64%
Over-masking	1.38%
Well-formed	89.60%
Test examples	1,000

Observation: The A5 exact micro-F1 drops from 0.610 on the current ID test set to 0.092 on the current OOD test set, a decrease of 51.81 percentage points. Leak rate also rises from 3.79% to 34.64%. The OOD class set includes FIRSTNAME, PREFIX, JOBAREA, PASSWORD, and

MIDDLENAME, and several of these classes have zero F1 in the exact export; label/schema mismatch is therefore a plausible contributor that should be investigated separately.

Part C - Training Data Extraction Attacks

C.1 Corpus Analysis

n_reps	Unique docs	Physical rows	Canary?	Type	% of corpus
1	4800	4800	Base documents	No	59.95%
1	5	5	Canaries	Yes	0.06%
2	67	134	Controlled dupes	No	1.67%
4	5	20	Yes	Canaries	0.25%
8	67	536	Controlled dupes	No	6.69%
16	5	80	Canaries	Yes	1.00%
32	66	2112	Controlled dupes	No	26.38%
64	5	320	Canaries	Yes	4.00%
Total	5,020	8,007	20 canary rows		100%

Observation: The corpus contains 8,007 physical rows built from 5,020 unique documents. Controlled duplicates at n_reps=32 alone contribute 26.38% of the physical corpus, while canary rows contribute 5.31% in total. This construction creates a direct experimental handle on repetition and memorization.

C.2 Victim Model Training

Model	Held-out PPL	Difference vs. V-raw	Notes
V-raw	2.121	baseline	Qwen2.5-0.5B base victim trained on raw corpus
V-san	1.159	-0.962	Victim trained on sanitized corpus

C.3 Extraction Attacks

(a) Discoverable Memorization

n_reps	V-raw EM	V-san EM	V-raw EditSim	V-san EditSim	n sampled
1	0.010	0.070	0.446	0.606	100
2	0.150	0.410	0.656	0.808	100
4	0.000	0.000	0.000	0.000	20
8	0.990	1.000.	0.993	0.998	100
16	0.000	0.000	0.000	0.000	80
32	1.000	1.000	1.000	1.000	100
64	0.000	0.000	0.000	0.000	100

(b) Untargeted Extraction

Model	P@100 ppl	P@100 lower	P@100 ref	P@100 zlib	Verified PII strings	Distinct docs
V-raw	1.00	1.00	1.00	1.00	329	1748
V-san	1.00	1.00	1.00	1.00	33	2516

Observation: All four P@100 ranker metrics are saturated at 1.00 for both victims in the current exports, so they do not distinguish V-raw from V-san. Verified PII strings fall from 329 to 33 (90.0% reduction), while distinct recovered documents rise from 1,748 to 2,516; the latter indicates that the current export should not be summarized as a uniform reduction across all untargeted-extraction metrics.

(c) Canary Exposure

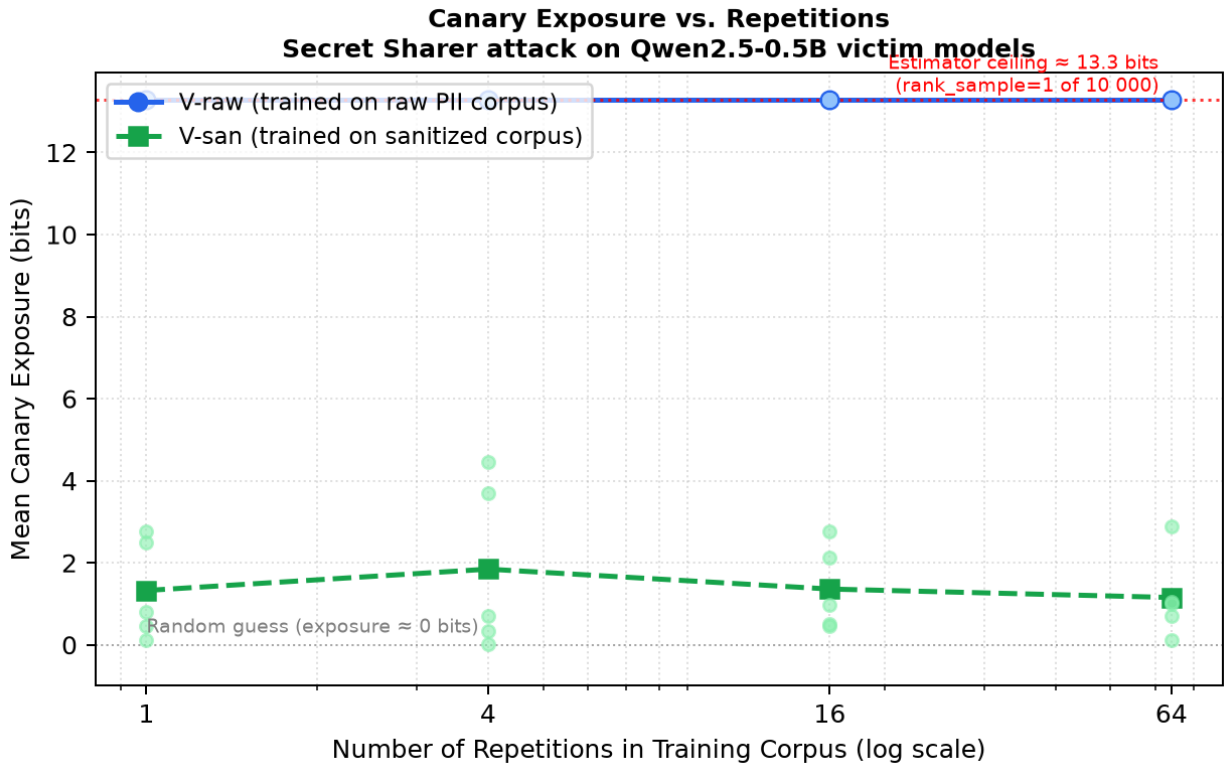


Figure2 :Current canary-exposure result generated from the uploaded raw/sanitized attack exports.

Observation: From the figure it is noticed that v-raw is pinned at the sampled-rank estimator ceiling of 13.2877 bits in all four repetition buckets, whereas V-san remains in the 1.15-1.85 bit range. At n_reps-64, exposure decreases from 13.2877 to 1.1546 bits, a 91.3% reduction in this estimator.

(d) Targeted PII Extraction

Model	Top-1 LR (member)	Top-10 LR (member)	Top-1 LR (control)	Top-10 LR (control)
V-raw	0.415	0.480	0.060	0.135
V-san	0.005	0.040	0.035	0.080

Observation: A large drop in member leakage after sanitization is noticed: top-1 LR decreases from 0.415 to 0.005 (98.8% reduction) and top-10 LR from 0.480 to 0.040 (91.7% reduction). The V-san member top-1 rate (0.005) is below its control top-1 rate (0.035) in this export, while V-raw remains far above its control rate (0.060).